

Storage Devices

There is a place in computer where all the data and information is saved so that can be used whenever it is needed. There two main category of storage device or memories

1. Primary/ Main memory
2. Secondary/Semi-Permanent/Auxiliary/Backup memory

Main Memory

Is also known as the internal memory of computer, it has the following types

1. RAM
2. ROM

RAM	ROM
Stand for RANDOM ACCESS MEMORY	Stands for Read Only Memory
Its temporary memory. It erases as soon as the computer is switched off	Permanent memory. Memory does not erase even if the computer is turned off
Known as Volatile memory	Known as non-volatile memory, also called as character generator
Used to store current running programs	Used to store programs that are not frequently used
Also known as read/write memory	Also known as BIOS Used in Booting process

Secondary Memory

Each computer has its own memory despite being so large it is still too small for storing large amount of data. Therefore to supplement the capacity of computer's internal memory secondary memories are used. There are two main types of secondary memory

1. Sequential Access
2. Direct Access

Sequential & Serial Access

It refers to read/write data in an order that is one after another. For instance if we want to read record 10 for this purpose we will have to read through 0 till 9 to read 10. Sequential memories are non-addressable memories

Difference between sequential and serial access is that in serial access data is read in the order its saved on the tape

e.g records are saved in the order 1, 5, 2, 4, 3 So in serial access after reading record 1 record 5 is read instead of 2 and so on. Where as in sequential access records are sorted in an order and then read. e.g ascending or descending order

Example of Sequential Access memory is **Magnetic Tape**(used to store backup of file server, can store upto 1tera bytes, Expensive as compare to CD's and DVD's.)

Direct Access

It is an addressable memory that. In direct access we do not have to read through all the records coming before the required record. We only have to mention the address of that record to access it directly. Direct access memories are faster than Sequential Access memories.

Example of Direct Access memory

- 1) *Magnetic Disc*
- 2) *Optical Disc*
- 3) *Solid state Storage*

Magnetic Disc

Magnetic disc are just like a phonograph record but in magnetic discs data is saved in the form of magnetic spots. The presence of a spot is measured as “1” and absence as “0”

Types of magnetic Discs

There are two types of magnetic disc

1. *Hard Disc*
2. *Floppy Disc*

Hard Disc	Floppy Disc
Can store huge amount of data up to 300GB	Can store up to 1.44MB
Gives faster access to data	Slower than Hard disk
Less portable	More portable than Hard disk
Durable (hard to break)	Fragile (breakable)
Hard disc has two platter with two read/write heads one for each side	It has single platter with one read/write heads

Optical Disc

These discs are made up of plastic. In optical discs data is saved in the form of pits or depressions rather than magnetic spots

Types of Optical disc

1. CD-ROM (Compact Disc Read Only Memory)

- Can store up to 700MB
- Faster than a floppy
- More durable than a floppy
- More portable than a hard disc
- Used to sell copies of music, movies, games, software etc
- Programmed when manufactured

2. CD-R (Compact Disc Recordable)

- Manufactured as blank memory
- It can be programmed once programmed cannot be erased

3. CD-RW (Compact Disc Re-writeable)

- It's a type of CD-R that can be erased, once erased it can be reprogrammed.
- Used to move files from one computer to another

4. DVD (Digital Versatile Disc)

- Can store up to 4.7GB
- Faster than a floppy and CD
- More durable than a floppy and CD
- More portable than a hard disc
- Used to sell copies of music, movies, games, software etc
- Programmed when manufactured

5. DVD-R (Digital Versatile Disc Recordable)

- Manufactured as blank memory
- It can be programmed once programmed cannot be erased

6. DVD-RW (Digital Versatile Disc Re-writeable)

- It's a type of DVD-R that can be erased, once erased it can be reprogrammed.
- Used to move files from one computer to another

7. Blue ray

- It can store upto 100GB
- More durable than a CD and DVD
- Used to store high definition images and Videos.

8. DVD-RAM

- Its faster than ordinary DVDs
- Can store more data than a DVD
- As in ordinary DVD's there is a single track that spirals around the whole Disc. DVD-RAM has individual Tracks
- We can read and write on DVD-RAM at the same time

Units of memory

Bit=1,0

1byte=8bits

1KB=1024bytes

1MB=1024KB

1GB=1024MB

1TB=1024GB

Solid state Storage

Two types

- Flash Memory (EEPROM)
electronically erasable programmable
read only memory
Used in mobile phones and digital
cameras
- Memory Stick

More portable than CD and Hard discs
Faster than CD and DVD
More Durable than CD and DVD